```
 GNU nano 8.7.1                memory_layout.c *
#include <stdio.h>
#include <stdlib.h>

// Global variable (Initialized) -> Data Segment
int global_initialized = 42;

// Global variable (Uninitialized) -> BSS Segment
int global_uninitialized;

// Function to demonstrate code segment (Text segment)
void sample_function() {
    printf("Code Segment (Text): Address of sample_function = >
}

int main() {
    // Static variable -> Data/BSS Segment
    static int static_var = 10;

    // Local variable -> Stack Segment
    int stack_var = 5;

    // Heap variable -> Heap Segment (allocated via malloc)
    int *heap_var = (int *)malloc(sizeof(int));
    if (heap_var == NULL) {
        perror("malloc failed");
        exit(1);
    }

    printf("=== Linux Process Address Space Layout ===\n\n");

    // 1. Code Segment
    sample_function();
    printf("Code Segment (Text): Address of main            = >
```

```c
    // 1. Code Segment
    sample_function();
    printf("Code Segment (Text): Address of main            = %p\n", (void*)main);

    // 2. Data Segment (Initialized Global & Static)
    printf("Data Segment       : Address of global_initialized = %p\n", (void*)&global_initialized);
    printf("Data Segment       : Address of static_var         = %p\n", (void*)&static_var);

    // 3. BSS Segment (Uninitialized Global)
    printf("BSS Segment        : Address of global_uninitialized = %p\n", (void*)&global_uninitialized);

    // 4. Heap Segment
    printf("Heap Segment       : Address of dynamically allocated = %p\n", (void*)heap_var);

    // 5. Stack Segment
    printf("Stack Segment      : Address of local stack_var    = %p\n", (void*)&stack_var);

    // Bonus: Inspecting /proc/self/maps for running process memory layout
    printf("\n=== Process Virtual Memory Maps (/proc/self/maps) ===\n");
    char command[64];
    snprintf(command, sizeof(command), "cat /proc/%d/maps", getpid());
    system(command);

    // Free allocated heap memory
    free(heap_var);

    return 0;
```

```
PS C:\Users\Gireesh> wsl
gireesh@Kannigantiii:/mnt/c/Users/Gireesh$ cd ~/oslab
gireesh@Kannigantiii:~/oslab$ nano memory_layout.c
gireesh@Kannigantiii:~/oslab$ gcc memory_layout.c -o memory_lay
out
memory_layout.c: In function ‘main’:
memory_layout.c:51:61: error: implicit declaration of function
‘getpid’ [-Wimplicit-function-declaration]
   51 | mand, sizeof(command), "cat /proc/%d/maps", getpid());
      |                                             ^~~~~~
gireesh@Kannigantiii:~/oslab$ gcc memory_layout.c -o memory_lay
out
./memory_layout
memory_layout.c: In function ‘main’:
memory_layout.c:51:61: error: implicit declaration of function
‘getpid’ [-Wimplicit-function-declaration]
   51 | mand, sizeof(command), "cat /proc/%d/maps", getpid());
      |                                             ^~~~~~
-bash: ./memory_layout: No such file or directory
gireesh@Kannigantiii:~/oslab$ nano memory_layout.c
gireesh@Kannigantiii:~/oslab$ gcc memory_layout.c -o memory_lay
out
gireesh@Kannigantiii:~/oslab$ ./memory_layout
```

```
=== Linux Process Address Space Layout ===

Code Segment (Text): Address of sample_function = 0x5f28cefc526
9
Code Segment (Text): Address of main           = 0x5f28cefc529
2

Data Segment       : Address of global_initialized = 0x5f28cefc
8010
Data Segment       : Address of static_var         = 0x5f28cefc
8014

BSS Segment        : Address of global_uninitialized = 0x5f28ce
fc801c

Heap Segment       : Address of dynamically allocated heap_var
= 0x5f28ed745010

Stack Segment      : Address of local stack_var    = 0x7ffd0d72
9c24

=== Process Virtual Memory Maps (/proc/self/maps) ===
5f28cefc4000-5f28cefc5000 r--p 00000000 08:30 16175
          /home/gireesh/oslab/memory_layout
5f28cefc5000-5f28cefc6000 r-xp 00001000 08:30 16175
          /home/gireesh/oslab/memory_layout
5f28cefc6000-5f28cefc7000 r--p 00002000 08:30 16175
          /home/gireesh/oslab/memory_layout
5f28cefc7000-5f28cefc8000 r--p 00002000 08:30 16175
          /home/gireesh/oslab/memory_layout
5f28cefc8000-5f28cefc9000 rw-p 00003000 08:30 16175
          /home/gireesh/oslab/memory_layout
5f28ed745000-5f28ed766000 rw-p 00000000 00:00 0
          [heap]
79116e200000-79116e228000 r--p 00000000 08:30 13614
          /usr/lib/x86_64-linux-gnu/libc.so.6
79116e228000-79116e3c0000 r-xp 00028000 08:30 13614
```